

1. In which phase do bacteria begin to form endospores or other survival structures due to nutrient depletion?

A. Stationary Phase

- b. Death (Decline) Phase
- c. Lag Phase
- d. Log (Exponential) Phase

Who is credited with the discovery of vaccination and the term “vaccine”?

- a. Shibasaburo Kitasato
- b. Edward Jenner
- c. Joseph Meister
- d. Louis Pasteur

Which of the following is an enveloped virus?

- a. Poliovirus
- b. SARS-CoV-2
- c. Human Papillomavirus
- d. All of the above

The Varicella Zoster Virus (VZV) is an example of

- a. Rotavirus
- b. Herpesvirus
- c. Parvovirus
- d. All of the above

Extracellular slime layers produced by bacteria are composed predominantly of:

- a. Proteins
- b. Complex polysaccharides
- c. Lipids
- d. Nucleic acids

Which of these organisms is a cestode?

- a. *Paragonimus westermani*
- b. Tapeworm
- c. *Fasciola hepatica*
- d. Guinea worm

The counter-stain in acid-fast staining is _

- a. Iodine
- b. Carbol fuchsin
- c. Acid-alcohol
- d. Methylene blue

Fimbriae are important for bacterial:

- a. Genetic material transfer
- b. Synthesis of flagella
- C. Adherence to surfaces
- d. Formation of biofilms

During which phase of the bacterial growth curve does the number of new cells produced equal the number of cells dying?

A. Log (Exponential) Phase

- b. Lag Phase
- C. Stationary Phase
- d. Death (Decline) Phase

Which of the following bacteria will give positive results for a coagulase test?

- a. *S. epidermidis*
- b. *S. saprophyticus*
- c. *S. hominins*
- d. *S. aureus*

Who was the first scientist to observe and describe microorganisms accurately?

- a. Anton Van Leeuwenhoek
- b. Robert Hooke
- c. Robert Koch
- d. Louis Pasteur

Which of the following is a psychrophile?

- a. *Pseudomonas fluorescens*
- O b. *E. coli*
- c. *Bacillus stearothermophilus*
- d. *Staphylococcus aureus*

Parasitism may refer to an association where

- a. Both populations benefit from the
- c. One population is unaffected while the other
- d. One population benefit while the other is

Association

- b. Both populations are harmed

Benefits

Harmed

Which of the following is a double-stranded DNA virus?

- a. Human papillomavirus
- b. SARS-CoV-2
- c. MERS-CoV
- d. Rhinovirus

Gram-positive bacteria are stained

Grain staining

During

- a. Violet

- b. Yellow
- c. Pink
- d. Purple

Which of the following is a naked virus?

- a. Ebola
- b. SARS-CoV-2
- c. Human Papillomavirus
- d. HIV

Microbiology involves the study of...

- a. Microorganisms using a microscope
- b. Molecules that are visible to human eyes
- c. Organisms that are not visible to the naked eye
- d. Animals and their family

Coccidioidomycosis is caused by

- a. Histoplasma
- b. C. immitis
- c. Cryptococcus
- d. C. immanitis

Why are germicides more effective at high temperatures?

- a. Molecules have higher kinetic energy, increasing the reaction rate

- b. High temperatures inhibit microbial growth
 - b. Microorganisms are less resistant at high temperatures
- d. Molecules are more stable at high temperatures

Which of the following is a Gram-positive bacterium responsible for causing food poisoning?

- a. *Staphylococcus aureus*
- b. *Escherichia coli*
- c. *Salmonella enterica*
- d. *Campylobacter jejuni*

Which of the following is/are positive for the catalase test?

- a. *C. diphtheria*
- b. *Pseudomonas*
- c. All of the above
- d. *Listeria*

All the following *Treponema* species cause diseases in humans except:

- a. *T. pertenue*
- b. *T. paraluisancuniculi*
- c. *T. carateum*
- d. *T. pallidum*

Which of the following is a sexually transmitted infection caused by a spirochete bacterium?

- a. Syphilis
- b. Gonorrhea
- c. Chlamydia
- d. Herpes

The destruction of all forms of microbial life, including bacterial spores is termed

- a. Sanitization
- b. Disinfection
- c. Sterilization
- d. Antisepsis

During which phase might bacterial infections produce the most severe symptoms due to toxin production?

- A. Log (Exponential) Phase
- b. Stationary Phase
- c. Death (Decline) Phase
- d. Lag Phase

Which hosts are involved in the life cycle of Plasmodium parasites?

- a. Insects and mosquitoes
- b. Humans and mosquitoes
- c. Mosquitoes only
- d. Birds and mosquitoes

Which of the following fungi is said to be opportunistic?

- a. Candida
- b. All of the above
- c. Cryptococcus
- d. Aspergillus

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What term describes the process where blood leukocytes engulf disease-causing bacteria?

- a. Phagocytosis
- b. Attenuation
- c. Pinocytosis
- d. Vaccination

Which condition can Campylobacter enteritis mimic?

- a. Typhoid fever
- b. Malaria
- c. Acute appendicitis
- d. Tuberculosis

The individual units of the viral capsid are known as

- a. Capsid protein
- b. Capsomers
- c. Tegument
- d. Envelope

Which of the following culture methods is best for the agglutination test?

- a. Stroke culture
- b. Lawn culture
- c. Stab culture
- d. Streak culture

Which type of bacteria has a thick peptidoglycan layer?

- a. Gram-negative bacteria
- b. Mycoplasmas
- c. Gram-positive bacteria
- d. Cyanobacteria

Which of the following is a virulence factor?

- a. Ability to adhere to a host
- b. Ability to colonize a host
- c. All of the above
- d. Ability to evade host defenses

What is the difference between disinfection and antisepsis?

- a. Disinfection is used for sanitization, while antisepsis is used for sterilization.
- b. Disinfection kills all microorganisms, while antisepsis kills only bacteria.
- c. Disinfection and antisepsis are the same

Process.

- d. Disinfection is used on inanimate objects, while antisepsis is used on living tissues.

Which measure(s) is/are most appropriate to prevent infections in burns patients?

- a. Practice stringent hand hygiene between patient contacts
- b. Use appropriate personal protective equipment (PPE)
- c. All of the above
- d. Care for patients in the burns unit with filtered air at positive pressure

Which of the following is a protozoan parasite causing malaria?

- a. Plasmodium vivax
- b. Trypanosoma cruzi
- c. Entamoeba histolytica
- d. Giardia lamblia

Capsid is associated with:

- a. Metazoan
- b. Fungi
- c. Bacteria

- d. Viruses

Which family do herpes viruses belong to?

- a. Flaviviridae
- b. Herpesviridae
- c. Adenoviridae
- d. Papillomaviridae

Which of the following is a Gram-positive bacterium?

- a. Bacillus
- b. Streptococcus
- c. Neisseria
- d. Staphylococcus

How are Salmonella serotypes typically classified?

- a. By their shape and arrangement
- b. By their biochemical characteristics
- c. By their motility pattern
- d. By their O and H antigens

The condition in which pathogenic microbes penetrate host defenses, enter tissues, and multiply is termed

- a. Pathogenicity
- b. Virulence
- c. Invasion
- d. Infection

Which of the following is a mesophile?

- a. *Pseudomonas fluorescens*
- b. *B. stearothermophilus*
- c. *E. coli*
- d. *Thermococcus*

What is a germicide?

- a. A chemical agent that only kills fungi
- b. A chemical agent that inhibits microbial growth
- d. A chemical agent that kills microorganisms
- d. A chemical agent that removes microorganisms physically

Which of the following is a true fungal pathogen?

- a. *Aspergillus*
- b. All of the above
- c. *Coccidioides*
- d. *Cryptococcus*

Which of the following is a true fungal pathogen?

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- b. All of the above
- c. Coccidioides
- d. Cryptococcus

The physical removal of microorganisms using soaps or detergents is termed

- a. Disinfection
- b. Sanitization
- c. Sterilization
- d. Degerming

What is the primary function of capsules in bacterial cells?

- a. Aid attachment to host cells and protect against host defenses
- b. Facilitate cell division
- c. Aid in respiration
- d. Protect against antibiotics

Which factors influence the susceptibility of microorganisms to germicides?

- a. All of the above
- b. Ability to form spores or cysts
- c. Presence or absence of a capsule
- d. Composition of cell wall

The capsid and the nucleic acid in the viral structure are referred to as

- a. Nucleocapsid
- b. Nucleoside
- c. nucleosome
- d. Envelope

Which of the following is a lactose fermenter?

- a. E. coli
- b. Proteus
- c. Salmonella
- d. Shigella

Who is known as the father of Microbiology?

- a. Ferdinand Cohn
- b. Robert Koch
- c. Anton van Leeuwenhoek
- d. Edwin John Butler

Which of the following is a Gram-positive bacterium?

- a. Staphylococcus
 - b.
- Neisseria

- b. Streptococcus
- d. Bacillus

In a mutualistic association

- a. Both organisms are harmed
- b. one organism benefits and the other is harmed
- b. One organism benefits and the other is unaffected
- d. both organisms benefit

What defines competition in a microbial population?

- a. Both populations are adversely affected concerning their survival and growth
- b. One population is unaffected while the other is harmed
- c. One population benefits while the other is harmed
- d. Both populations benefit from the association

A true pathogen is a microorganism

- a. Capable of causing disease in healthy persons with normal immune defenses
- b. A temporary microorganism
- c. A microorganism that cannot cause disease
- d. Capable of causing disease only in immunocompromised hosts

What characterizes amensalism in bacterial-host interaction?

- a. One population benefits while the other is harmed

- b. Both populations are adversely affected
- c. One population produces substances that inhibit another population
- d. Both populations benefit from the association

Which of the following is used as a mordant in Gram staining?

- a. Alcohol
- b. Safranin
- c. Crystal violet
- d. Iodine

Which phase is characterized by no apparent cell division but high metabolic activity as bacteria adapt to a new environment?

- A. Log (Exponential) Phase
- b. Stationary Phase
- c. Death (Decline) Phase
- d. Lag Phase

All the following form part of Koch's postulate except:

- a. The suspected microorganism must be isolated and grown in a pure culture.
- b. The microorganism must be present in some cases of the disease but absent from healthy organisms.
- c. The same disease must result when the isolated microorganism is inoculated into a healthy host.
- d. The same microorganism must be isolated again from the diseased host.

What is the main killing agent when microwaves are used to control microorganisms?

- a. Lon formation
- b. The heat generated by water molecule vibration
- c. Thymine dimer formation
- d. Radiation

Which of the following has a capsule around the cell wall?

- a. *Klebsiella pneumonia*
- b. All of the above
- b. *Neisseria*
- d. *Vibrio cholerae*

A chemical agent that prevents the multiplication of bacteria without necessarily killing them may refer to

- a. Bactericide
- b. Fungicide
- c. Germicide
- d. Bacteriostatic agent

What is the primary function of capsules in bacterial cells?

- a. Protect against antibiotics
- b. Facilitate cell division

- c. Aid in respiration
- d. Aid attachment to host cells and protect against host defenses

Bacteria with flagella clusters at both cell poles are known as?

- a. Lophotrichous
- b. Amphitrichous
- c. Monotrichous
- d. Peritrichous

HIV-1 belong to the family of:

- a. Parvoviridae
- b. Poxviridae
- c. Reoviridae
- d. Retroviridae

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- c. Reoviridae
- d. Retroviridae

What is the effect of ultraviolet light on the DNA of microorganisms?

- a. It converts DNA into RNA
- b. It causes DNA to replicate uncontrollably
- c. It breaks the DNA strands
- d. It induces thymine dimers

Hospital-acquired (nosocomial) infections are said to be

- a. Infections that are present at birth
- b. Infections acquired through foodborne transmission
- c. Diseases acquired during a hospital stay
- d. Infections acquired outside of a hospital

All the following are gram-positive spherical bacteria, except

- a. E. coli.
- b. S. epidermidis
- c. S. aureus
- d. E. faecalis

Which of the following is a bacterial toxin that causes muscle paralysis by blocking acetylcholine release at neuromuscular junctions?

- A. Tetanus toxin
- b. Cholera toxin
- c. Diphtheria toxin
- d. Botulinum toxin

Which of the following culture methods is best for bacteriophage typing?

- a. Stab culture
- b. Streak culture
- c. Stroke culture
- d. Lawn culture

Which of the following is indole positive?

- a. Klebsiella
- b. Salmonella
- c. E. coli
- d. All of the above

Which of the following areas are most affected by nosocomial infections?

- a. Mouth, throat, and ears
- b. Eyes, nose, and mouth
- c. Skin, hair, and nails

- d. Urinary tract, respiratory tract, and surgical incisions

Which of the following is a viral vector vaccine?

- a. AstraZeneca
- b. Sinopharm
- c. Pfizer
- d. Sinovac

Which family do herpes viruses belong to?

- a. Flaviviridae
- b. Adenoviridae
- c. Herpesviridae
- d. Papillomaviridae

Which of the following culture methods is best for demonstrating gelatin liquefaction or hydrolysis test?

- A. Stab culture
- b. Stroke culture
- c. Lawn culture
- d. Streak culture

Histoplasmosis is a condition caused by

- a. Cryptococcus
- b. Histoplasma neoformans

O c. Aspergillus

- d. Histoplasma capsulatum

What must occur for disease to happen in the presence of normal flora?

- a. Lower temperature
- b. Increased humidity
- c. Increased nutrient availability
- d. A change in the body's environment

Which of the following is positive for the urease test?

- a. Klebsiella
- b. Salmonella
- c. a and c
- d. Proteus

Which of the following is a sexually transmitted infection caused by a spirochete bacterium?

- a. Chlamydia
- b. Syphilis
- c. Herpes
- d. Gonorrhea

Which of the following is positive for the oxidase test?

- a. Shigella
- b. All of the above
- c. Salmonella
- d. Pseudomonas